Module 1 — Read 0

Project preview — Project 1 — Testing a Claim

Module 1 Preview

Big question: Is what we observed **surprising**?

> **Note**
>
> This page is a **short orientation**, not the textbook. Read before Read 1 so you see how Project 1 fits the course story (five steps, three worlds, and "theory before data").

Good science: state your claim before you observe

In Module 0, Read 0 you met the **five-step statistical process**. Project 1 is your first full pass through it.

Step	In Project 1
1 — Ask	A clear research question about a **proportion** p (probability of success on one trial)
2 — Plan	Define p, write H_0 and H_a, decide how you will analyze and conclude — **before** you collect all $n = 25$ trials (milestone)
3 — Collect	Observe the process; record x successes out of n
4 — Analyze	Describe what you saw; compare x to the **null model** you planned under H_0; compute a **p-value**
5 — Conclude	Interpret the p-value in context on a one-page report and make a conclusion about the answer to your research question based on the data you collected and analyzed.

Best practice: Step 2 comes **before** Step 3. Good scientists do not pick H_0 after seeing the data x. Instead they state the claim they want to test, build the imagined model, *then* gather data. This best practice is reinforced by doing Phase A before Phase B on the project page and in the milestone notebook.

Three worlds — in the right order

Statistics moves among three worlds (Module 0). For Project 1, the order matters:

1. The world we imagine (theory first)

Here you work with claims and models, not your dataset yet.

- Turn your research question into H_0 and H_a with theorized value for p in H_0.
- Pretend H_0 is true: if the null proportion were correct, what counts X of successes in n trials would be typical?
- That pretend world is a binomial model: fixed n, fixed p from H_0, independent binary trials.

Your milestone null plot (step A5) is a picture of this imagined world, drawn before you know your observed value x.

2. The world we observe (observations second)

Only after the plan do you collect data.

- Run $n = 25$ trials from a process that satisfies the **four BRP conditions**.
- Record $x =$ number of successes and find $\hat{p} = x/n$.

- Descriptive charts (bar or pie) describe what happened, they do not test the claim by themselves.

3. The world of computers (connect the two)

The programming language R help us move between what we imagine and what we observe:

- **`dbinom`** + **`geom_col`** compute bar heights and show them for $P(X = x)$ under H_0
- **`geom_vline`** — mark your observed x on that null plot
- **`pbinom`** — p-value summarizes how unusual your observed x is if the H_0 you imagined were actually true, relative to the H_a you imagined.

You will take a deep dive into the mathematics of the theoretical world in Read 1 and how to use it to test your observation in Read 2.

Observation meets theory: the p-value

Once you have planned the model and collected x, you ask:

> If the world I imagined under H_0 were true, how often would I see a count as or more extreme than my observed x (in the direction of H_a)?

That probability is the p-value.

- A **small** p-value means your x would be **unusual** if H_0 were true — the observed data and the null claim do not fit well.
- A **large** p-value means your x is **plausible** under H_0 — the data are consistent with the null model (not proof that H_0 is true).

> **!** One mistake to avoid early
>
> The **p-value is not** the probability that H_0 is true. It is the probability of data **at least this extreme**, *assuming* H_0 is true.

A deep dive

Most inferential methods in later modules rely on approximations or software you will be asked to use based on trust. Project 1 is different: the binomial test is one of the few cases where, with high-school algebra and patience, you can understand the whole story, from assumptions through the probability formula to the p-value.

We will build it up in Read 1 and Read 2:

1. Probability language (events, p, independence).
2. The binomial pmf — exact probability for each possible count x.
3. The four BRP conditions — when that formula applies to your real process.
4. Hypotheses, null plots, and p-values in context.

At the end of the module you will use just one line of code (**`pbinom()`** in R) to compute a p-value but you will have a deep understand of exactly what this line is computing, how to interpret it and when to use it.

The binomial probability formula (preview)

Let X = number of successes in n independent trials, each with the same probability p of success. Under that model,

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n$$

Read this as: choose which x of the n trials are successes ($\binom{n}{x}$ ways), multiply success probability p on those x trials and failure probability $(1-p)$ on the remaining $n-x$ trials.

For Project 1 you plug in $n = 25$ and p from H_0 to draw the **null distribution**. Your observed x is a single bar on that picture; the **p-value** is the total probability in the tail(s) at least as extreme as x toward H_a.

Four BRP conditions

Your data collection must satisfy a **binomial random process** (BRP). You will defend all four in your report; check them in context.

Condition	Meaning (in your study)
(a) Binary	Each trial has exactly two outcomes (success / failure)
(b) Independent	One trial does not change the chance of success on another
(c) Fixed p	The probability of success is the same on every trial (under H_0 for the model)
(d) Fixed n	You decide the number of trials in advance (Project 1 uses $n = 25$)

If any condition fails, the binomial formula (and `pbinom`)do not describe your process honestly.

What you will do in practice

Phase	You understand…	You compute with…
Plan (milestone)	Question, p, H_0/H_a, BRP, null plot	`dbinom` + `geom_col` (no x yet)
Collect	One real x from your process	Your own data
Analyze & report	Descriptive $\hat{p}$; tail probability under H_0	`geom_vline`, `pbinom`, one-page write-up

Tour the sample project

Open the **sample one-page write-up — ESP binomial test (PDF)**. It is a finished Project 1 report.

Section	What to notice	World
Introduction	Question, motivation, p in context, H_0 and H_a stated before data	Imagine
Data collection	How, when, where; **four BRP conditions in context**	Observe
Descriptive statistics	x, n, $\hat{p}$, chart of observed counts	Observe
Inferential statistics	Null binomial plot, line at x, p-value from `pbinom`	Imagine + computer
Conclusion	What the p-value means for the question (not "accept H_0")	Imagine (but more accurate)

What you will do (your project, not ESP)

By the end of Module 1 you will:

1. Choose your **own** binomial process ($n = 25$), not ESP or other course examples.
2. Complete the **milestone** — theory and null plot **before** full data collection.
3. Collect data, then analyze with R and write a **one-page** report (rubric).